The arithmetic-geometric mean of Gauss (9.15)

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The arithmetic-geometric mean of Gauss. Let $0 < a < b$. We define the two sequences a_n and b_n as follows.

$$a_0 = a, \quad b_0 = b, \qquad a_{n+1} = \sqrt{a_n b_n}, \quad b_{n+1} = \frac{a_n + b_n}{2}.$$

(a) Prove that $a_n < a_{n+1}$, $b_n > b_{n+1}$ and $a_n < b_n$ for all n.
(b) Prove that $b_{n+1} - a_{n+1} = (b_n - a_n)^2/8b_{n+2}$.
(c) Show that $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n = g$ with a quadratic convergence rate.